

Multi-Source Candidate Data Transformer — Technical Design

Eightfold Engineering Intern Assignment (Jul–Dec 2026) | [Shalu Kumari](#) | shalujha6044@gmail.com

1. Pipeline

detect → **extract** → **normalize** → **match/cluster** → **merge+confidence** → **project** → **validate** → **emit**

- **detect**: Sniffs file extension and content header; routes to the correct adapter. Unreadable files are logged and skipped, never fatal.
- **extract**: Produces `RawRecord` field values tagged with `source_id`, `source_type`, and `method`. Only layer aware of source-specific shapes.
- **normalize**: Per-field normalizers (phones, dates, country, skills) run independently of source. Failures close to `null` + provenance note — never invented.
- **match/cluster**: Union-Find over exact email → exact phone → conservative name+company. Near-duplicates (same name+company, conflicting email/phone) are flagged via `possible_duplicate_of` rather than auto-merged.
- **merge+confidence**: `source_trust × method_confidence` picks the winner per field. Every losing value is preserved in provenance — conflicts are made visible, not silently resolved.
- **project** → **validate** → **emit**: A read-only projection layer reshapes the immutable canonical record per the runtime config, validates types and required fields, then emits JSON.

2. Canonical Schema & Normalized Formats

Fields: `candidate_id` (opaque UUID hex; stable within a run, not across runs), `full_name`, `emails[]`, `phones[]`, `location{city,region,country}`, `links{}`, `headline`, `years_experience`, `skills[{name,confidence,sources[]}]`, `experience[]`, `education[]`, `provenance[]`, `overall_confidence`, `possible_duplicate_of[]`.

- **phones** → E.164 (phonenumbers). Configurable default region. Ambiguous → `null` + `normalize_failed: ambiguous_region`.
- **dates** → YYYY-MM. Year-only → YYYY-01 with `precision: year` provenance note. No fabricated months.
- **country** → ISO-3166 alpha-2 via name/alias lookup. **skills** → lowercased + alias-mapped (e.g. 'ReactJS' → `react`). Unknowns kept as-is with lower confidence.
- **Scale**: Extraction/normalization are per-record. Clustering is in-memory Union-Find $O(n \alpha(n))$. No DB needed; fits comfortably on thousands of candidates.

3. Matching, Merge Policy & Confidence

Source type	Trust	Method	Confidence
recruiter_csv	0.90	direct_field	1.00
ats_json	0.85	regex_extract	0.70
resume	0.75	heuristic_parse	0.65
recruiter_notes	0.55	derived	0.80

Agreeing independent sources earn a corroboration boost (+0.05/source, capped 1.0). Highest-confidence value is emitted; all losers stay in

`provenance.overall_confidence` = mean over populated fields only (missing fields don't penalize; a wrong guess would).

4. Runtime Output Config

The `CanonicalRecord` is immutable. A separate **projection layer** reads it through the config: field subset, path remapping (`emails[0]`, `skills[].name`), per-field normalization override, provenance/confidence toggles, and an `on_missing` policy (`null` / `omit` / `error`). Config paths are validated at **load time** — a typo fails fast rather than silently emitting `null` at runtime.

5. Edge Cases & Scope

- **Near-duplicate identity**: Same name+company, conflicting emails → two separate profiles cross-referenced via `possible_duplicate_of` (confidence 0.4, `fuzzy_name_company_match`). Human reviews; pipeline doesn't guess.
- **Stated vs. computed experience**: Date-math (`derived`, weight 0.80) can outrank a stated heuristic value (0.65). Both preserved in provenance.
- **Ambiguous phone**: Unresolvable country code → `null` + `normalize_failed: ambiguous_region`. Never guessed.
- **Corrupt file**: Extractor logs a `WARNING`, contributes zero fields, rest of pipeline continues unaffected.
- **Config path typo**: Rejected at load time with a clear error, not silently `null` at runtime.

Deliberately out of scope: Live GitHub/LinkedIn APIs (auth/ToS overhead; CSV+ATS+resume+notes satisfy structured+unstructured requirement); fuzzy name-only matching (high false-positive risk for hiring decisions); UI beyond the CLI (per the assignment's own stated priorities).